

Precision-dependent invariance costs and the geometry of initial sets

Abstract

We study the number of open-loop controls required to keep every actual initial state near a constraint set when the tolerance decreases with the time horizon. For a planar linear family whose feasible branches have different contraction rates, we determine the cost for every compact initial set with interior. We also construct compact initial sets with area arbitrarily close to that of the constraint and a strictly smaller cost, although their exact and outer invariance rates are unchanged. The lower bounds apply to arbitrary approximate controls, including trajectories that leave the constraint set. A separate theorem for common radial dynamics gives one precision law for all positive-area compact initial sets, allowing nonlinear transverse terms. Finally, a fixed example with an expanding branch is resolved by cyclically selected symbolic witnesses. The results distinguish the geometry of actual controlled feasibility from the classical symbolic counting used to evaluate its cost.

Keywords. Invariance entropy; finite-horizon control; vanishing tolerance; contraction; initial-set geometry.

1 Introduction

Invariance entropy counts the controls needed to preserve a state constraint. Its exact version requires membership in the constraint set at every time, whereas its outer version first fixes a positive tolerance and then lets the time horizon increase. These two requirements can have different asymptotic costs. A joint limit, in which the tolerance decreases as the horizon grows, asks a more quantitative question: how much of the exact feasibility information remains necessary at a specified ambient resolution?

We answer this question for two classes of planar controlled systems. The main result concerns two linear maps preserving a forward cone by an appropriate choice of input. Both maps are contractions in a common norm, but their contraction factors differ. For every compact initial set with interior, the cost is determined by the largest symbolic entropy attainable before contraction reaches the prescribed tolerance. In the same system, compact initial sets with area arbitrarily close to that of the whole cone can have a strictly smaller cost. Their exact invariance rate is nevertheless the same, and their ordinary outer rate is still zero. Thus nearly full area does not determine the joint precision–horizon cost.

The distinction is between worst-case feasible contraction along all symbolic prefixes and the uniform typical contraction on a large compact subset. Neither symbolic calculation alone gives a lower bound for the control problem: an approximate control need not follow the exactly feasible branch. We therefore estimate the actual states that an arbitrary control can cover after its first incorrect branch choice. For initial sets with interior, one control covers only linearly many witnesses from an appropriate stopped tree. For large compact sets of uniformly typical points, the same first deviation gives a sectional measure bound. These are the steps that connect classical counting to the ambient constrained dynamics.

For comparison, we prove a different theorem with common radial dynamics and a strict inward control margin. It allows nonlinear transverse terms and gives the same cost for every positive-area compact initial set. Its assumptions are not contained in those of the binary linear family, or conversely. The comparison identifies two distinct mechanisms without claiming a general dichotomy. A final example with one expanding branch shows how to recover a matching lower bound when the stopped-interval argument no longer applies.

Relation to previous work

The control-catalogue formulation and the distinction between exact and outer invariance are standard; see Kawan [8], especially Sections 2.1 and 3.2. Volume growth and control selection already enter invariance-entropy estimates. For hyperbolic control sets, Da Silva and Kawan [4] express the cost through unstable volume growth under their control-set hypotheses. Our forward cone does not satisfy those hypotheses: a positive radial coordinate only decreases, so states cannot approximately reach larger radii within the cone. We do not obtain a new formula for hyperbolic control sets.

Precision requirements also occur in entropy for exponential stabilization [1], practical stabilization [2], and state estimation [9]. In stabilization the bound is imposed relative to a target and changes with the current time. In state estimation it bounds the discrepancy from a reference trajectory. Here the initial state is never approximated: a selected control must keep its actual trajectory within one tolerance of a fixed set throughout the entire horizon. Only between horizons does that tolerance change. The results below concern this catalogue problem, not an encoder–decoder implementation theorem.

Yao’s preprint [13] gives a continuous-time construction with a Cantor matching constraint, positive exact cost and zero approximate cost. Its matching and fading estimates already show why late control disagreements may disappear at fixed ambient resolution. The mere separation of exact and outer rates is consequently not our contribution. Our main assertion is the explicit joint limit and its strict dependence on initial-set geometry in a positive-area planar constraint, including compact initial sets of nearly full area.

The symbolic tools have a separate, established history. Stopping at a product scale and entropy-to-contraction ratios occur in self-similar geometry [7]; the binomial estimates used here are elementary instances of the method of types [3]. Dimension constructions for control entropy were developed by Huang and Zhong [6], and Bowen equations for invariance pressure on partitions by Yang et al. [12]. Measure-theoretic invariance entropy and variational principles are studied in [11, 10]. We do not claim a new dimension equation or pressure duality. In particular, partition-based pressure fixes an invariant partition, whereas our infimum runs over all actual controls satisfying an ambient tolerance. Proposition 3.1 supplies the comparison with symbolic covers in the contracting binary family; Section 5 gives the separate estimate on nearly full-area sets. Neither comparison is assumed in the symbolic calculation. Likewise, the minimum-partial-sum rotation in the last example is a classical cyclic argument; see [5].

2 The catalogue problem and the main result

Let U be a finite alphabet and let $f_i : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $i \in U$, be continuous maps. All sequences $u = (u_1, u_2, \dots) \in U^{\mathbb{N}}$ are allowed. Put $\varphi(0, x, u) = x$ and $\varphi(k, x, u) = f_{u_k} \circ \dots \circ f_{u_1}(x)$ for $k \geq 1$. Throughout, the constraint and metric are

$$Q = \{(s, z) : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{dist} = \text{dist}_2. \quad (2.1)$$

In particular, $\text{area}(Q) = 1$. Write $Q^\delta = \{x : \text{dist}_2(x, Q) < \delta\}$.

Definition 2.1. For compact nonempty $K \subset Q$, integer $n \geq 1$, and $\delta > 0$, let $r(n, \delta; K)$ be the least cardinality of a finite set $\mathcal{C} \subset U^{\mathbb{N}}$ such that

$$\forall x \in K \quad \exists u \in \mathcal{C} \quad \forall k \in \{0, \dots, n\} : \quad \varphi(k, x, u) \in Q^\delta. \quad (2.2)$$

Let $r_E(n; K)$ be the corresponding exact count with Q^δ replaced by Q . Set both counts equal to one at $n = 0$, and set a count equal to $+\infty$ if no finite catalogue exists. The outer rate is

$$h_{\text{out}}(K) = \lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \ln r(n, \delta; K).$$

All logarithms are natural. We evaluate the joint limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln r(n, \delta_n; K), \quad \delta_n > 0, \quad \delta_n \rightarrow 0, \quad -\frac{1}{n} \ln \delta_n \rightarrow \gamma \in [0, \infty]. \quad (2.3)$$

No monotonicity of δ_n is assumed.

Only the first n input symbols affect a horizon n . There are $n + 1$ constrained states, including the actual initial state. A word meeting (2.2) may leave Q ; it need not agree with any particular exactly feasible coding.

Our main family is

$$F_i(s, z) = (\rho_i s, \rho_i(2z + v_i s)), \quad v_0 = 1, \quad v_1 = -1, \quad 0 < \rho_0, \rho_1 < \frac{1}{2}. \quad (2.4)$$

These are smooth linear diffeomorphisms. Define

$$\begin{aligned} \alpha &= \ln 2, \quad c_i = -\ln \rho_i, \quad \bar{c} = (c_0 + c_1)/2, \quad c_{\min} = \min(c_0, c_1), \quad c_{\max} = \max(c_0, c_1), \\ H(p) &= -p \ln p - (1-p) \ln(1-p), \quad \chi(p) = p c_0 + (1-p) c_1, \quad 0 \ln 0 = 0, \\ \mathcal{F}(\gamma) &= \max_{0 \leq p \leq 1} H(p) \min\{1, \gamma/\chi(p)\}, \quad \mathcal{G}(\gamma) = \alpha \min\{1, \gamma/\bar{c}\}. \end{aligned}$$

At $\gamma = \infty$ the minima are interpreted as one.

Theorem 2.2 (Dependence on the initial set). *For the family (2.4), the following hold.*

- (i) *Q is controlled invariant. Every trajectory from a bounded set converges to the origin uniformly over all input sequences. For every compact $K \subset Q$ with positive area,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln r_E(n; K) = \alpha, \quad h_{\text{out}}(K) = 0.$$

- (ii) *For every compact $K \subset Q$ with nonempty interior in $\mathbb{R}^2$, and every tolerance sequence satisfying (2.3),*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln r(n, \delta_n; K) = \mathcal{F}(\gamma). \quad (2.5)$$

- (iii) *For every $\varepsilon \in (0, 1)$ there is a compact $K_\varepsilon \subset Q$ with $\text{area}(K_\varepsilon) > 1 - \varepsilon$ such that, for every tolerance sequence satisfying (2.3),*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln r(n, \delta_n; K_\varepsilon) = \mathcal{G}(\gamma). \quad (2.6)$$

The same K_ε works for all γ and all such sequences.

- (iv) *If $\rho_0 \neq \rho_1$, then $\mathcal{F}(\gamma) > \mathcal{G}(\gamma)$ for $0 < \gamma \leq c_{\min}$.*

This theorem does not classify arbitrary positive-area compact sets. The sets in part (iii) have empty interior. They show that even arbitrarily small loss of area can change the rate, while part (ii) ensures that every set containing an open subset retains the full rate.

Example 2.3. Take $\rho_0 = 1/4$, $\rho_1 = 1/16$, and $\delta_n = 2^{-n}$. Writing $\phi = (1+\sqrt{5})/2$, Theorem 2.2 gives

$$\mathcal{F}(\ln 2) = \frac{1}{2} \ln \phi > \frac{1}{3} \ln 2 = \mathcal{G}(\ln 2).$$

Both classes of initial sets have exact rate $\ln 2$ and outer rate zero. The computation and the strict inequality are justified in Section 5.3.

3 Feasible words and arbitrary approximate controls

For a finite word $w = w_1 \cdots w_k$, put

$$A_w = \prod_{j=1}^k \rho_{w_j}, \quad S(w) = -\ln A_w, \quad A_\emptyset = 1.$$

For $s > 0$ use $y = z/s$ only as a calculation coordinate. Under F_i the ratio becomes $2y + v_i$, and the radius after w is sA_w . The inverse ratio branches

$$b_0(y) = (y - 1)/2, \quad b_1(y) = (y + 1)/2$$

map $[-1, 1]$ onto its two halves. Thus

$$I_w = b_{w_1} \circ \cdots \circ b_{w_k}([-1, 1]), \quad |I_w| = 2^{1-k}, \quad (3.1)$$

is precisely the set of ratios exactly feasible through the word w . The intervals at each depth cover $[-1, 1]$ with disjoint interiors. Shared endpoints have no effect on any count or measure estimate below. This proves controlled invariance, including at the fixed origin.

Choose $M \geq 1$ so large that

$$L = \rho_{\max}(2 + 1/M) < 1, \quad \|(s, z)\|_M = \max\{M|s|, |z|\}, \quad \rho_{\max} = \max_i \rho_i. \quad (3.2)$$

Each F_i has operator norm at most L in this norm, and $\|x\|_2 \leq \sqrt{2} \|x\|_M$. This proves the uniform convergence claim in Theorem 2.2(i). A length- n exact word covers a cone slice of area $\int_0^1 s |I_w| ds = 2^{-n}$. Consequently

$$\text{area}(K) 2^n \leq r_E(n; K) \leq 2^n \quad (3.3)$$

for positive-area $K \subset Q$. For fixed $\delta > 0$, choose a fixed m with $M\rho_{\max}^m < \delta/(2\sqrt{2})$. Follow an exactly feasible prefix of length m and append any fixed tail. Every later state has Euclidean norm less than $\delta/2$. The catalogue has at most 2^m elements for all $n \geq m$, proving the outer assertion.

3.1 Comparison with a stopped tree

For $0 < t < 1$, stop every binary path at the first word w satisfying $A_w \leq t$, or at depth n if no earlier prefix meets this condition. Let $\mathcal{L}(n, t)$ be the complete prefix-free leaf set and $N(n, t) = |\mathcal{L}(n, t)|$. Write w^- for the word obtained by deleting the last symbol of w . For every leaf,

$$A_{w^-} > t, \quad A_w > \rho_{\min} t, \quad \rho_{\min} = \min_i \rho_i. \quad (3.4)$$

A leaf of depth less than n also has $A_w \leq t$.

Proposition 3.1 (Actual controls versus stopped words). *There are constants $B, C_0 > 0$, depending only on the maps, such that for all $n \geq 1$ and all sufficiently small $\delta > 0$,*

$$\frac{N(n, \delta)}{1 + Bn} \leq r(n, \delta; Q) \leq N(n, \delta/C_0). \quad (3.5)$$

If compact $K \subset Q$ has interior, there are a radius $s_ > 0$ and a fixed word a of length d such that, for $n > d$ and sufficiently small δ ,*

$$\frac{N(n - d, \delta/(s_* A_a))}{1 + Bn} \leq r(n, \delta; K) \leq N(n, \delta/C_0). \quad (3.6)$$

Proof. Set $C_0 = 2\sqrt{2}M$. For each leaf of $\mathcal{L}(n, \delta/C_0)$, use that word followed by a fixed tail. A leaf reaching n is exactly feasible for its cylinder throughout the horizon. At an earlier stopping time the endpoint has M -norm at most $M\delta/C_0$; by (3.2) its whole tail stays within Euclidean distance $\delta/2$ of the origin. This proves both upper bounds.

For the lower bound fix $s_* \in (0, 1]$, take $t = \delta/s_* < 1$, and associate to each leaf its midpoint witness $(s_*, s_* y_w)$, where y_w is the midpoint of I_w . A necessary condition for $(r, z) \in Q^\delta$ with $r \geq 0$ is

$$|z| < r + 2\delta. \quad (3.7)$$

Indeed a point $(r', z') \in Q$ within distance δ has $|r - r'| < \delta$ and $|z - z'| < \delta$.

Consider any input word u of length n that covers some of these witnesses in the sense of (2.2). At most one leaf is a prefix of u . For any other covered witness let $j \leq |w|$ be the first position where $w_j \neq u_j$, and put $a' = u_1 \cdots u_{j-1}$. If $m_{a'}$ is the midpoint of $I_{a'}$, applying the wrong branch gives

$$|y_j| - 1 = 2^j |y_w - m_{a'}|.$$

Its actual radius is $s_* A_{a'} \rho_{u_j}$. Hence (3.7) confines y_w to the opposite child in a one-sided band of width at most

$$b_j = \frac{2\delta}{s_* 2^j A_{a'} \rho_{\min}}. \quad (3.8)$$

Every leaf in that child has

$$|I_w| = |I_{a'}| 2^{-(|w| - |a'|)} \geq |I_{a'}| \frac{A_w}{A_{a'}} > \frac{2^{2-j} \rho_{\min} \delta}{s_* A_{a'}} =: \ell_j. \quad (3.9)$$

The first inequality uses $\rho_i < 1/2$, and the last one uses (3.4). Midpoints of disjoint-interior intervals of length at least ℓ_j are separated by at least ℓ_j . Thus the band (3.8) contains at most $1 + b_j/\ell_j \leq 1 + (2\rho_{\min}^2)^{-1}$ such midpoints. Summing over j shows that one arbitrary control covers at most $1 + Bn$ witnesses. Taking $s_* = 1$ proves (3.5).

If K has interior, a ball inside K contains a fixed-radius segment $\{(s_*, s_* y) : y \in I_a\}$ for some dyadic interval I_a . For $\delta/s_* < A_a$, the leaves below a in $\mathcal{L}(n, \delta/s_*)$ are a concatenated with $\mathcal{L}(n - d, \delta/(s_* A_a))$. Their witnesses lie in K . The preceding bound still applies if u first disagrees within a . This proves (3.6). $\square$

The lower bound uses an error at an intermediate time. It imposes no condition on the approximate trajectory after that time, and in particular does not presume that approximate controls follow the exact coding.

4 Counting stopped words

Lemma 4.1. *For integers $k \geq 1$ and $0 \leq j \leq k$,*

$$\frac{e^{kH(j/k)}}{k+1} \leq \binom{k}{j} \leq e^{kH(j/k)}.$$

Proof. For $0 < j < k$, set $p = j/k$. The probability of j successes in a binomial distribution with parameters k, p is $\binom{k}{j}e^{-kH(p)}$. It is at most one and is a mode, as the ratios of adjacent probabilities show. A largest one of $k+1$ probabilities is at least $1/(k+1)$. The endpoint cases are immediate. $\square$

Proposition 4.2. *If $t_n \rightarrow 0$ and $-n^{-1} \ln t_n \rightarrow \gamma \in [0, \infty]$, then*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln N(n, t_n) = \mathcal{F}(\gamma).$$

Proof. Put $T_n = -\ln t_n$. A word of length k with j zeros has cost $k\chi(j/k)$. An early leaf satisfies

$$T_n \leq k\chi(j/k) < T_n + c_{\max};$$

a terminal leaf of length n satisfies $n\chi(j/n) < T_n + c_{\max}$. Suppose first that $0 < \gamma < \infty$. There are at most $(n+1)^2$ length–type pairs. Bound the number of leaves of each pair by Lemma 4.1, allowing all words of that type. Along a subsequence maximizing the exponential count, pass to limits $k/n \rightarrow a$ and $j/k \rightarrow p$. For early leaves $a\chi(p) = \gamma$, and the exponent is at most $aH(p)$. For terminal leaves $a = 1$ and $\chi(p) \leq \gamma$, so it is at most $H(p)$. Both are bounded by $\mathcal{F}(\gamma)$; the polynomial number of types does not change the exponential rate.

Conversely fix $p \in [0, 1]$ and $0 < a < \min\{1, \gamma/\chi(p)\}$. Choose $k_n = \lfloor an \rfloor$ and $j_n/k_n \rightarrow p$. All words of this type have cost less than T_n for large n . Costs are positive, so no prefix has stopped. Distinct words have disjoint descendant leaf sets, and therefore $N(n, t_n) \geq \binom{k_n}{j_n}$. Lemma 4.1 gives a lower rate $aH(p)$. Increase a to the indicated minimum and maximize over p .

If $\gamma = 0$, every path stops by depth $\lceil T_n/c_{\min} \rceil + 1$, unless it reaches the earlier horizon. The number of leaves is bounded by $2^{\min(n, \lceil T_n/c_{\min} \rceil + 1)}$, which has zero rate. If $\gamma = \infty$, eventually $T_n > nc_{\max}$, and every path reaches depth n ; then $N(n, t_n) = 2^n$. $\square$

Fixed multiplicative changes in t_n and fixed shifts of the horizon do not alter this limit. Proposition 3.1 now proves Theorem 2.2(ii).

5 Nearly full area with a smaller cost

5.1 A compact set of uniformly typical ratios

For a non-dyadic $y \in (-1, 1)$, let $i_1(y)i_2(y)\cdots$ be its unique feasible coding. Under normalized Lebesgue measure on $[-1, 1]$ these digits are independent and uniformly distributed, since every length- k cylinder has measure 2^{-k} . Hence

$$\frac{S_k(y)}{k} := \frac{1}{k} \sum_{j=1}^k c_{i_j(y)} \longrightarrow \bar{c} \quad \text{for almost every } y. \quad (5.1)$$

For completeness, the moment-generating function of the number of zeros, together with $\cosh t \leq e^{t^2/2}$, gives a probability at most $2e^{-2k\eta^2}$ for a frequency deviation exceeding η . Borel–Cantelli for $\eta = 1/m$, $m \geq 1$, proves (5.1).

Egorov's theorem and inner regularity give compact sets $E \subset (-1, 1)$, excluding all dyadic points, of measure as close to 2 as desired, on which (5.1) is uniform. For every $\zeta > 0$ there is $C_\zeta \geq 1$ such that

$$C_\zeta^{-1} e^{-(\bar{c}+\zeta)k} \leq A_k(y) \leq C_\zeta e^{-(\bar{c}-\zeta)k} \quad (y \in E, k \geq 0). \quad (5.2)$$

Indeed uniform convergence gives these bounds for large k , and a single constant handles the finitely many earlier times.

Choose $s_0 \in (0, 1)$ and set

$$K_E = \{(s, sy) : s_0 \leq s \leq 1, y \in E\}. \quad (5.3)$$

It is compact and has area $(1 - s_0^2)|E|/2$, since the coordinate map $(s, y) \mapsto (s, sy)$ has Jacobian s . Taking $s_0^2 < \varepsilon/2$ and $|E| > 2(1 - \varepsilon/2)$ gives $\text{area}(K_E) > 1 - \varepsilon$. The set has empty interior because every ratio interval contains a dyadic point absent from E .

5.2 Matching estimates for arbitrary controls

Fix $0 < \zeta < \bar{c} - \alpha$. Let $T = -\ln \delta$. For the upper bound use the natural prefixes meeting E at length

$$m = \min \left\{ n, \left\lceil \frac{T + \ln(2\sqrt{2}MC_\zeta)}{\bar{c} - \zeta} \right\rceil \right\}. \quad (5.4)$$

There are at most 2^m of them. If $m < n$, the endpoint has M -norm at most $MC_\zeta e^{-(\bar{c}-\zeta)m} \leq \delta/(2\sqrt{2})$. Append any fixed tail and use (3.2). If $m = n$, use the exact catalogue. Thus the upper rate is at most $\alpha \min\{1, \gamma/(\bar{c} - \zeta)\}$, with the usual convention at infinity.

For the lower bound consider any control u , any fixed $s \in [s_0, 1]$, and any $m \leq n$. The ratios in E that agree with u through time m lie in one interval of length 2^{1-m} . If the first disagreement is at $j \leq m$, the common prefix satisfies (5.2). The same first-deviation calculation as in (3.8) bounds the measure of those ratios by

$$C\delta 2^{-j} e^{(\bar{c}+\zeta)(j-1)},$$

where C depends on s_0, ζ and the maps, but not on u, s, j, n or δ . Summing this geometric progression, using $\bar{c} + \zeta > \alpha$, shows that the set of ratios covered by u has measure at most

$$C'(2^{-m} + \delta e^{(\bar{c}+\zeta-\alpha)m}). \quad (5.5)$$

Fubini's theorem and the Jacobian $s \leq 1$ give the same bound, up to a constant, for the area of points of K_E covered by one control.

For sufficiently small δ choose $m = \min\{n, \lceil T/(\bar{c} + \zeta) \rceil\}$. Since $\delta e^{(\bar{c}+\zeta)m} \leq 1$, the second term in (5.5) is at most 2^{-m} . Positive area then gives $r(n, \delta; K_E) \geq b_\zeta 2^m$ for some $b_\zeta > 0$. Applied to (2.3), this gives the lower rate $\alpha \min\{1, \gamma/(\bar{c} + \zeta)\}$. Letting $\zeta \downarrow 0$ in both estimates proves (2.6), including $\gamma = 0$ and $\gamma = \infty$. The construction of E preceded the choice of ζ and of the tolerance sequence, so the required quantifier order holds.

5.3 Strict separation

Let $D > 0$ be the unique solution of

$$\rho_0^D + \rho_1^D = 1. \quad (5.6)$$

Set $p_D = \rho_0^D$. The nonnegativity of relative entropy gives

$$H(p) - D\chi(p) = -p \ln \frac{p}{p_D} - (1-p) \ln \frac{1-p}{1-p_D} \leq 0,$$

with equality at $p = p_D$. Consequently $\max_p H(p)/\chi(p) = D$, and for $0 \leq \gamma \leq c_{\min}$,

$$\mathcal{F}(\gamma) = D\gamma, \quad \mathcal{G}(\gamma) = \frac{\alpha}{c}\gamma.$$

If $c_0 \neq c_1$, strict arithmetic-geometric mean at $D_0 = \alpha/\bar{c}$ gives $e^{-D_0 c_0} + e^{-D_0 c_1} > 2e^{-D_0 \bar{c}} = 1$. The left side decreases in D , so $D > D_0$. This proves Theorem 2.2(iv) and completes its proof.

In Example 2.3, setting $x = 4^{-D}$ in (5.6) gives $x + x^2 = 1$, whence $D = \ln \phi / (2 \ln 2)$ and $\bar{c} = 3 \ln 2$. The inequality $\phi^3 = 2\phi + 1 > 4$ proves the claimed strict gap. The equation (5.6) is classical; its role here is to evaluate a rate whose comparison with actual controls has already been proved.

6 A common radial law for all positive-area sets

We now consider a separate class. Let $q > 1$ and let $V \subset \mathbb{R}$ be a finite nonempty set for which there exists $\eta > 0$ satisfying

$$\forall y \in [-1, 1] \quad \exists v \in V : \quad |qy + v| \leq 1 - 2\eta. \quad (6.1)$$

Let $R : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing C^∞ diffeomorphism such that

$$R(0) = 0, \quad 0 < R(s) < s \quad (0 < s \leq 1), \quad \rho = R'(0) \in (0, 1]. \quad (6.2)$$

For each $v \in V$ let $P_v : \mathbb{R}^2 \rightarrow \mathbb{R}$ be smooth, with $P_v(0) = 0$ and $DP_v(0) = 0$, and assume

$$\kappa := \max_{v \in V} \sup_{x \in \mathbb{R}^2} \max\{|P_v(x)|, \|DP_v(x)\|, \|D^2 P_v(x)\|\} < \min\{\eta, q/2\}. \quad (6.3)$$

Derivative norms in this hypothesis are Euclidean operator norms. Define

$$f_v(s, z) = (R(s), t(s) [qz + sv + P_v(s, z)]), \quad t(s) = \begin{cases} R(s)/s, & s \neq 0, \\ \rho, & s = 0. \end{cases} \quad (6.4)$$

Use Definition 2.1 with these maps, the same Q , and all sequences in $V^\mathbb{N}$ allowed.

Theorem 6.1 (Common radial precision law). *Under (6.1)–(6.3), the maps (6.4) are smooth diffeomorphisms and Q is controlled invariant. For every compact positive-area $K \subset Q$ and every tolerance sequence satisfying (2.3), the limit is*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln r(n, \delta_n; K) = \Psi_{\alpha, c}(\gamma) := \max_{0 \leq \alpha \leq 1} \{\alpha\gamma - (ac - \gamma)_+\}, \quad \alpha = \ln q, \quad c = -\ln \rho. \quad (6.5)$$

Here $x_+ = \max\{x, 0\}$. For finite γ , this equals

$$\Psi_{\alpha, c}(\gamma) = \begin{cases} \alpha, & c = 0, \\ \min\{\alpha, \alpha - c + \gamma\}, & 0 < c \leq \alpha, \\ \alpha \min\{1, \gamma/c\}, & c > \alpha. \end{cases} \quad (6.6)$$

At $\gamma = \infty$ it equals α . Moreover $\lim_n n^{-1} \ln r_E(n; K) = \alpha$.

6.1 Viability and an exact prefix catalogue

The identity $t(s) = \int_0^1 R'(\theta s) d\theta$ shows that t is smooth and positive. For fixed s , the vertical map is strictly increasing, with derivative $t(s)(q + \partial_z P_v) > 0$, and is proper because P_v is bounded. Together with invertibility of R , the inverse function theorem proves that each f_v is a global smooth diffeomorphism.

Taylor's formula and (6.3) give

$$|P_v(s, sy)| \leq \kappa s^2 \quad (|y| \leq 1), \quad |\partial_z P_v(x)| \leq \min\{\kappa, \kappa \|x\|_2\}. \quad (6.7)$$

For $s > 0$ the ratio evolves as

$$y_{j+1} = qy_j + v_{j+1} + P_{v_{j+1}}(s_j, s_j y_j)/s_j, \quad s_j = R^j(s).$$

The last term has absolute value at most $\kappa s_j \leq \kappa < \eta$ in Q . The inward margin (6.1) therefore proves viability with margin at least η . The origin is fixed.

Since R is increasing, $R^j(s) \leq R^j(1) \rightarrow 0$ uniformly on $s \in [0, 1]$: the decreasing limit for $s = 1$ is a fixed point, and (6.2) excludes every positive fixed point in this interval. For each $\xi \in (0, \rho)$ there is $C_\xi > 0$ such that

$$\frac{R^j(s)}{s} \leq C_\xi(\rho + \xi)^j \quad (0 < s \leq 1, j \geq 0). \quad (6.8)$$

For each $s_0 > 0$ there is also $d_{\xi, s_0} > 0$ such that

$$\frac{R^j(s)}{s} \geq d_{\xi, s_0}(\rho - \xi)^j \quad (s_0 \leq s \leq 1, j \geq 0). \quad (6.9)$$

To see this, after a common finite time all radii are in a neighborhood where $\rho - \xi \leq R(s)/s \leq \rho + \xi$. The finitely many earlier products have a uniform upper bound, and a positive lower bound on $[s_0, 1]$.

Lemma 6.2. *For every $e > 0$ there is $D_e \geq 1$ such that $r_E(m; Q) \leq D_e e^{(\alpha+e)m}$ for every $m \geq 0$.*

Proof. Fix a block length N . Cover $[-1, 1]$ by a mesh of at most $C_\eta q^N$ centers, so that each ratio is within $\eta q^{-N}/4$ of a center. For each center choose a length- N word for the unperturbed ratio system $y \mapsto qy + v$ that keeps every subsequent ratio in $[-1 + 2\eta, 1 - 2\eta]$; this is possible by (6.1).

Choose $a_N > 0$ so small that $\kappa a_N \sum_{j=0}^{N-1} q^j \leq \eta/4$. For any $0 < s \leq a_N$, apply the center's word to the actual system starting from a nearby ratio. As long as the actual ratios lie in $[-1, 1]$, their discrepancy from the reference at each time $k \leq N$ is bounded by

$$q^k \frac{\eta q^{-N}}{4} + \kappa a_N \sum_{j=0}^{k-1} q^j \leq \eta/2.$$

Induction using the reference margin establishes the required membership in $[-1, 1]$ at every step, and hence justifies this estimate throughout the block. This single block catalogue works for every $s \leq a_N$.

A common finite time m_0 brings every feasible radius below a_N . Use all length- m_0 words, which include a feasible choice for each initial state, and thereafter concatenate the block catalogues. A remaining block of length less than N can use all words of that length. For fixed N , this gives exact upper rate at most $\ln q + N^{-1} \ln C_\eta$. Choose N large enough for the extra term to be less than e and absorb the finite initial and remaining blocks into D_e . $\square$

6.2 A sectional lower bound at every intermediate time

Fix $s \in (0, 1]$ and a word u . Write $Z_k(z)$ for the vertical coordinate at time k when the initial state is (s, z) . Each Z_k is strictly increasing. The set $J = J(n, \delta, s, u)$ of $z \in [-s, s]$ whose whole trajectory through time n belongs to Q^δ is an interval, possibly empty. Indeed, Q^δ is convex; its section at the fixed radius $R^k(s)$ is an interval; and its inverse under Z_k is an interval. Intersect these intervals over $k = 0, \dots, n$.

The chain rule gives

$$Z'_k(z) = \frac{R^k(s)}{s} \prod_{j=0}^{k-1} (q + \partial_z P_{u_{j+1}}(s_j, Z_j(z))). \quad (6.10)$$

Given $\zeta \in (0, q)$, choose a small $d > 0$ and a common time N_ζ so that $s_j \leq d$ for $j \geq N_\zeta$ and $\kappa\sqrt{10}d \leq \zeta$. If $\delta < d$ and $z \in J$, then (3.7) implies $|Z_j(z)| < s_j + 2\delta \leq 3d$ at those times. Thus (6.7) bounds each late derivative factor below by $q - \zeta$. The early factors are at least $q - \kappa > 0$. After decreasing a constant to cover all early values of k , one obtains

$$Z'_k(z) \geq A_\zeta \frac{R^k(s)}{s} (q - \zeta)^k \quad (z \in J, 0 \leq k \leq n), \quad (6.11)$$

uniformly in n, u, s and sufficiently small δ .

The interval $Z_k(J)$ has length at most $2R^k(s) + 4\delta$. Applying the mean value theorem on subintervals of J , then taking their increasing union, gives from (6.11) and (6.9), for $s \in [s_0, 1]$,

$$|J| \leq C(q - \zeta)^{-k} (1 + C'\delta(\rho - \xi)^{-k}), \quad 0 \leq k \leq n. \quad (6.12)$$

The constants are independent of the control and the horizon. This is a bound for controls satisfying the entire time interval, tested at any chosen time k .

If K has positive area, choose $s_0 > 0$ so that $K \cap \{s \geq s_0\}$ still has positive area. By Fubini, every spanning catalogue satisfies

$$r(n, \delta; K) \geq \frac{B(q - \zeta)^k}{1 + C'\delta(\rho - \xi)^{-k}} \quad (0 \leq k \leq n). \quad (6.13)$$

For finite γ in (2.3), take $k = \lfloor an \rfloor$. The logarithm of the denominator divided by n tends to $(a[-\ln(\rho - \xi)] - \gamma)_+$. First let $n \rightarrow \infty$, then $\zeta, \xi \downarrow 0$, and finally maximize over $a \in [0, 1]$. This proves the lower bound in (6.5). When $\gamma = \infty$, take $k = n$ to obtain the lower rate α . For the exact count the same argument with $\delta = 0$ applies directly to the exact feasible interval. It gives lower rate α , matching Lemma 6.2.

6.3 A feasible prefix followed by a full approximate tail

Suppose first that $\rho < 1$. Fix $a_r \in (\rho, 1)$. Equation (6.8) implies that the endpoint of any exactly feasible prefix of length m satisfies

$$\|x_m\|_M \leq C a_r^m, \quad (6.14)$$

where $\|(s, z)\|_M = \max\{M|s|, |z|\}$ and C is uniform in the prefix and the initial state in Q . At the origin,

$$Df_v(0) = \begin{pmatrix} \rho & 0 \\ \rho v & \rho q \end{pmatrix}.$$

For any $L > \rho q$, choosing M sufficiently large makes the operator norm of every such matrix less than L . Continuity and finiteness of V give a common convex neighborhood of the origin on which every f_v is L -Lipschitz in this norm. All maps fix the origin.

Append one fixed input tail after an exact prefix. Inductively, as long as it stays in the neighborhood, $\|x_k\|_M \leq C a_r^m L^{k-m}$ for $m \leq k \leq n$. For small enough δ , making this bound at most $\delta/(2\sqrt{2})$ for every k both proves that the induction remains in the neighborhood and ensures membership in Q^δ throughout the tail.

If $L > 1$, it is enough to take an exact prefix of length

$$m = \min \left\{ n, \max \left\{ m_0, \left\lceil \frac{n \ln L + \ln(2\sqrt{2}C/\delta)}{\ln L - \ln a_r} \right\rceil \right\} \right\}, \quad (6.15)$$

where m_0 is any fixed initial time needed for the local estimate. If the minimum selects n , the exact catalogue itself suffices. Otherwise the displayed inequality proves the complete tail bound. Lemma 6.2 gives the upper rate

$$(\alpha + e) \min \left\{ 1, \frac{\gamma + \ln L}{\ln L - \ln a_r} \right\}.$$

If $L \leq 1$, the maximum along the tail occurs at m , and one can instead take

$$m = \min \left\{ n, \max \left\{ m_0, \left\lceil \frac{\ln(2\sqrt{2}C/\delta)}{-\ln a_r} \right\rceil \right\} \right\}. \quad (6.16)$$

The resulting upper rate is $(\alpha + e) \min\{1, \gamma/(-\ln a_r)\}$.

Let $e \downarrow 0$ and $a_r \downarrow \rho$. If $\rho q > 1$, let $L \downarrow \rho q$ in the first bound. It gives $\min\{\alpha, \alpha - c + \gamma\}$. The same limit from $L > 1$ handles $\rho q = 1$. If $\rho q < 1$, choose $L < 1$ and use the second bound to get $\alpha \min\{1, \gamma/c\}$. For $\rho = 1$, the exact catalogue already gives upper rate α , which matches the lower bound. The case $\gamma = \infty$ is likewise bounded by the exact catalogue. A direct maximization over a gives (6.6), completing Theorem 6.1.

Example 6.3 (Why the terminal time is insufficient). Let $q = 2$, $V = \{-3/2, 0, 3/2\}$, $\eta = 1/8$, and $R(s) = s/4$. The margin (6.1) holds since every number in $[-2, 2]$ is within $3/4$ of one of $-3/2, 0, 3/2$. Set $P_v(s, z) = e_v \tanh s \tanh z$, with $|e_v| < 1/64$. These functions vanish to first order at the origin, and their function, gradient and Hessian norms are bounded by $4|e_v| < 1/16$. For every positive-area compact $K \subset Q$, Theorem 6.1 therefore gives rate $\min\{\ln 2, \gamma/2\}$. At $\delta_n = 2^{-n}$, the rate is $\frac{1}{2} \ln 2$. In (6.5) the maximizing time fraction is $a = 1/2$; testing only $a = 1$ would give zero. The whole time interval is essential.

Remark 6.4 (The two classes have different hypotheses). Theorem 6.1 has a strict inward margin and a common radial map; Theorem 2.2 has two nonoverlapping ratio branches and control-dependent radial factors. The latter alphabet has no strict inward margin at the branch boundary. Thus neither theorem is stated as an extension of the other. When $\rho_0 \neq \rho_1$, their difference also cannot be removed by a fixed C^1 state change taking origin to origin and respecting labels. The eigenvalues of $DF_i(0)$ are $\rho_i, 2\rho_i$, so its spectral radius is $2\rho_i$; in (6.4) every derivative has the same spectral radius ρq . A C^1 conjugacy makes corresponding derivatives similar, which is incompatible with two different spectral radii. No assertion about an arbitrary topological change of coordinates is needed here.

7 One expanding branch at the critical tolerance

The geometric inequality (3.9) uses $\rho_i \leq 1/2$. We finish with a fixed system for which it fails:

$$F_0(s, z) = \left(\frac{3s}{4}, \frac{3(2z + s)}{4} \right), \quad F_1(s, z) = \left(\frac{s}{4}, \frac{2z - s}{4} \right). \quad (7.1)$$

The exact ratio coding (3.1) remains valid. The map F_0 has transverse eigenvalue $3/2$, whereas F_1 contracts in the maximum norm with operator norm at most $3/4$.

Proposition 7.1. For (7.1), $K = Q$, and $\delta_n = 2^{-n}$, put $p_* = \ln 2 / \ln 3$. Then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln r(n, 2^{-n}; Q) = H(p_*). \quad (7.2)$$

More precisely, with $\lambda = \ln(p_*/(1 - p_*)) / \ln 3 > 0$ and the stopped-tree count for the factors $3/4, 1/4$,

$$\frac{\binom{n}{\lceil p_* n \rceil}}{n(3n+1)} \leq r(n, 2^{-n}; Q) \leq N \left(n, \frac{2^{-n}}{2\sqrt{2}} \right) \leq (8\sqrt{2})^\lambda e^{nH(p_*)}. \quad (7.3)$$

In particular, the displayed stopped catalogue exceeds the minimal actual catalogue by at most a factor Cn^3 for a fixed C .

Proof. For the upper bound, use each stopped prefix exactly, then append 1^∞ . At an early stopping time the endpoint has maximum norm at most $2^{-n}/(2\sqrt{2})$, and each tail step multiplies that norm by at most $3/4$. Thus its Euclidean norm stays at most $2^{-n}/2$. Terminal leaves are exactly feasible throughout.

To bound the leaves, write $c_0 = \ln(4/3)$, $c_1 = \ln 4$, and $T = n \ln 2 + \ln(2\sqrt{2})$. Every leaf w of length $k \leq n$ has $S(w) < T + c_1$. Give symbol 0 Bernoulli probability p_* . The mass of its cylinder is

$$\nu[w] = e^{-\beta k - \lambda S(w)}, \quad \beta = -\ln(1 - p_*) - \lambda c_1. \quad (7.4)$$

Here $1/2 < p_* < 3/4$. The function $g(p) = c_0 \ln(1 - p) - c_1 \ln p$ is strictly decreasing and $g(3/4) = 0$, so $\beta = g(p_*) / \ln 3 > 0$. The leaves form a complete prefix-free family whose masses sum to one. By (7.4), each mass is at least $e^{-\beta n - \lambda(T + c_1)}$. It follows that

$$N \leq e^{\beta n + \lambda(T + c_1)} = (8\sqrt{2})^\lambda e^{nH(p_*)},$$

because $p_* c_0 + (1 - p_*) c_1 = \ln 2$ and $\beta + \lambda \ln 2 = H(p_*)$.

For the lower bound, let $j_n = \lceil p_* n \rceil$ and consider words of length n with j_n zeros. For a word w define

$$D_k(w) = \ln(2^k A_{w_1 \dots w_k}) = N_0(w_1 \dots w_k) \ln 3 - k \ln 2.$$

Here $N_0(a)$ counts the zeros in a . Their total satisfies $0 \leq D_n(w) < \ln 3$. Call a word good if $D_k(w) \geq 0$ for every $0 \leq k \leq n$. Every cyclic orbit of these words contains a good one: start immediately after a minimum of $D_0, \dots, D_{n-1}$. Before wrapping, all new partial sums are differences from that minimum; after wrapping they are such differences plus the nonnegative total D_n . Each cyclic orbit has at most n elements, so there are at least $n^{-1} \binom{n}{j_n}$ good words. This is the elementary minimum-partial-sum rotation; no assertion about an exact number of good rotations is required.

For every good word choose the actual state $(1, y_w)$, where y_w is the midpoint of I_w . All such midpoints belong to a grid of spacing $2^{1-n} = 2\delta_n$. Fix any length- n input word u covering some of these states. One witness can have $w = u$. For any other one let j be the first disagreement and $a = u_1 \dots u_{j-1}$ the common prefix. As in (3.8), the wrong branch and (3.7) imply

$$|y_w - m_a| < \frac{\delta_n}{\rho_{u_j} e^{D_{j-1}(w)}} \leq 4\delta_n.$$

For each fixed j , this is a one-sided band, so it contains at most three points of the grid. Thus an arbitrary approximate control covers at most $3n + 1$ good witnesses. Dividing their count by $3n + 1$ proves the first inequality in (7.3).

By Lemma 4.1, $\binom{n}{j_n} \geq e^{nH(j_n/n)} / (n + 1)$. Since $j_n/n = p_* + O(1/n)$ and $p_* \in (0, 1)$, $nH(j_n/n) = nH(p_*) + O(1)$. The two bounds therefore give (7.2) and the stated polynomial ratio. $\square$

Unlike (2.4), this example does not have convergence for every input sequence. In fact

$$F_0^k(1, 1) = ((3/4)^k, 2(3/2)^k - (3/4)^k),$$

whose second coordinate diverges. Exactly feasible trajectories still converge to the origin, since $s_k \leq (3/4)^k$ and $|z_k| \leq s_k$. Proposition 7.1 addresses one specified tolerance and initial set. It is not a precision spectrum for general expanding branches or an extension preserving convergence under all inputs.

8 Scope of the results

The main distinction is visible within a single system: positive area and even nearly full area need not substitute for interior when contraction is tied to the feasible control branch. The common radial theorem provides an independent regime where positive area does suffice. The exact and outer rates alone do not record this distinction.

All estimates use the actual state projection and the Euclidean constraint neighborhood, and all upper bounds control the whole tail. The proofs rely on special one-dimensional ratio geometry. They do not classify all positive-area initial sets, establish a general formula for nonlinear control-dependent contraction, or produce a communication protocol. The cyclic witnesses resolve the particular expanding example without removing these restrictions. These limitations separate the proved control-geometric statements from the classical counting methods used to calculate their exponents.

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